MEMORANDUM

TO: Dr. Tina Han

FROM: Hengeltraub, Isabella; Wu, Xintong; Jhanwar, Yuvraj; Hassan, Khaleel

CC: ME32301 Lab Section 08- Harbin, Zachary; Alshaikh, Leen

DATE: April-11-2025

RE: Helical Extension Springs: Lab 9

EXECUTIVE SUMMARY

The primary goal of this lab is to determine shear modulus G and initial tension F_i of an extension spring under tensile loading. The shear modulus G of a material, sometimes referred to as Modulus of rigidity, is the measure of rigidity of a body [1]. Initial tension F_i is the force that holds the closely-wound springs together, which must be overcome before the spring deflects [2]. Three helical music-wire steel extension springs of various dimensions were tested during this lab. Table 1 shows the dimensions and testing details of each spring.

Material	D [mm]	OD [mm]	d [mm]	L_o [mm]	G [MPa]	F_i [N]	N_b	N_a	k [N/m]
Music-Wire Steel	6	7	1	25	80924	2.32	14	14.4	3.26
Music-Wire Steel	4.7	5.5	0.8	39	79793	3.32	38	38.4	1.03
Music-Wire Steel	7.3	8.5	1.2	41	80656	0.99	23	23.4	2.30

Table 1: Table of calculated values.

Spring dimensions such as coil diameter OD, wire diameter d, and free length L_0 were provided and mean coil diameter D was computed using

$$D = OD - d \tag{1}$$

At the start of the experiment, the spring was properly mounted onto the Instron UTM by securing the keyrings of the spring within the Instron UTM grips. Before the test was initiated, the force and displacement scales were zeroed. The deflection of the head of the UTM was used to map

spring deflection. Upon completing the test, the data was saved and exported and the specimen was removed. From Castigliano's second theorem and also taking initial tension F_i into account, the force-deflection relationship of an extension spring is given by

$$F = F_i + k\Delta \tag{2}$$

where Δ is spring deflection and the spring rate k referred to as the amount of force needed per unit of deflection [2] is given by

$$k = \frac{d^4 G}{8D^3 N_a} \tag{3}$$

Na, the number of active coils, is computed using

$$N_a = N_b + \frac{1}{2(1+\nu)} \tag{4}$$

and the number of body coils N_b is derived from the free length L_0 of the spring

$$L_0 = 2(D-d) + (N_b+1)d \tag{5}$$

Since the F-Δ has a non-linear relationship with G and F_i, a least squares regression model method was performed on MATLAB. Subsequently, F_i and G were obtained. The obtained G_{exp} was then compared to the literature value for each specimen and%diff was computed. As shown below 2, % diff across all three specimens was minimal, with the highest recorded value being just 1.15%. These values fall well within the acceptable range. Potential sources of error include calculations based on nonlinear regions of the graph, measurement inaccuracies, unit inconsistencies, misalignment and inadequate securement of the spring.

Specimen	G_{exp} [MPa]	G_{theo} [MPa]	% diff
25mm	80923.74	80000	1.15
39mm	79792.92	80000	0.26
41mm	80656.36	80000	0.82

Table 2: Percent difference between calculated and literature values of shear modulus.

RESULTS

In the table below, the measured deflection Δ is recorded for the 25mm specimen with force F from 0 N to 11.12 N in increments of 1.11 N.

F [N]	Δ [mm]
0	0
1.11	0.0479
2.22	0.1026
3.34	0.3109
4.45	0.6537
5.56	0.9964
6.67	1.3343
7.78	1.6794
8.90	2.0239
10.01	2.3654
11.12	2.6962

Table 3: Measured deflection for 25mm specimen.

In the table below, the measured deflection Δ is recorded for the 39mm specimen with force F from 0 N to 7.12 N in increments of 0.71 N.

F [N]	Δ [mm]
0	0
0.71	0.0368
1.42	0.0662
2.14	0.0970
2.85	0.1348
3.56	0.2759
4.27	0.9296
4.98	1.6243
5.69	2.3065
6.41	3.0089
7.12	3.6765

Table 4: Measured deflection for 39mm specimen.

In the table below, the measured deflection Δ is recorded for the 41mm specimen with force F from 0 N to 15.57 N in increments of 1.56 N.

F [N]	Δ [mm]
0	0
1.56	0.2580
3.11	0.9210
4.67	1.6004
6.23	2.2718
7.78	2.9482
9.34	3.6319
10.90	4.3036
12.46	4.9849
14.01	5.6602
15.57	6.3358

Table 5: Measured deflection for 41mm specimen.

In Figure 1, the relationship between force and deflection for each specimen is plotted, along with the least-squares models. The 25mm spring has the steepest slope, meaning its deflection increases the least as force increases. On the other hand, the 39mm spring has the least steep slope, meaning its deflection increases the most as force increases. The 41mm specimen has by far the largest range that corresponds to the linear trend, going from a displacement of around 0.2 mm to 6.3 mm, while the 39mm specimen has a range of around 0.2 mm to 3.7 mm and the 25mm specimen has a range of around 0.2 mm to 2.7 mm.

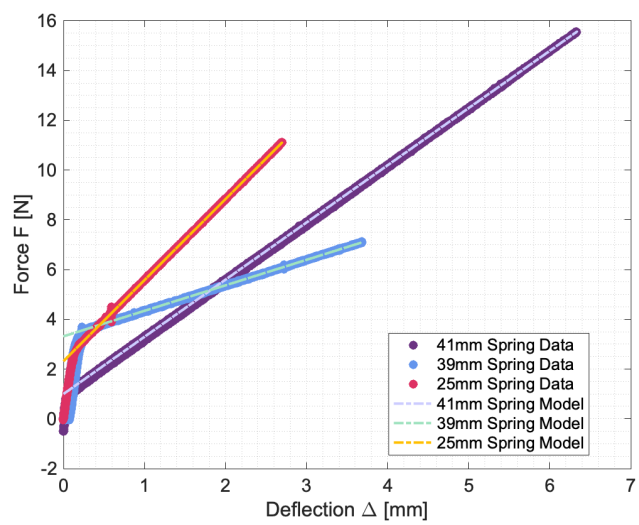

Figure 1: Force vs. deflection for all springs

Table 6 contains the calculated values for each specimen, with D being mean coil diameter, OD

being the outside coil diameter, d being wire diameter, L_0 being the free length of the spring, G being the shear modulus, F_i being the initial tension, N_b being the number of body coils, N_a being the number of body coils, and k being the spring rate.

The calculations were done using MATLAB code. The first file that was run was the "Cutoff Plots" code, which graphed the 25mm spring raw data, 39mm spring raw data, and 41mm spring raw data in individual plots. From this, the linear portions of each graph could be seen and their bounds were visually estimated. Next, a function was run for each of the raw data sets, which used the least-squares to calculate the shear modulus G and initial tension F_i of the helical extension spring, using the force and deflection data measured during the spring experiments. The next file that was run was the "Slope Calculations" file, which calculated the number of body coils N_b using Equation (5), number of active coils N_a based off the N_b using Equation (4), and then used N_a as well as the G from the function codes to find the spring rate k for each spring using Equation (3). Finally, a master plot with all three of the raw data sets was created, which also included a linear model for each of these data sets. The linear model was created as an extension of the linear section of each raw data set, using the F_i value of the least-squares model as the y-intercept, and the spring rate k from the "Slope Calculations" code as the slope.

The mean coil diameter D is calculated using Equation (1). The number of body coils N_b is calculated using Equation (5), the number of active coils N_a is calculated using Equation (4), and the spring rate k is calculated using Equation (3).

Material	D [mm]	OD [mm]	d [mm]	L_o [mm]	G [MPa]	F_i [N]	N_b	N_a	k [N/m]
Music-Wire Steel	6	7	1	25	80924	2.32	14	14.4	3.26
Music-Wire Steel	4.7	5.5	0.8	39	79793	3.32	38	38.4	1.03
Music-Wire Steel	7.3	8.5	1.2	41	80656	0.99	23	23.4	2.30

Table 6: Table of calculated values.

As seen in the table above, the shear modulus and spring rate is the lowest for the 39mm specimen and highest for the 25mm specimen, which matches the steepness of the slopes in Figure 1. Additionally, the initial tension is lowest for the 41mm specimen and highest for the 39mm specimen,

which matches the y-intercepts in the figure.

The calculated shear modulus is compared to the literature values of steel music wire in the table below [3], with G_{exp} being the calculated value, G_{theo} being the literature value, and % diff being the percent difference:

Specimen	G_{exp} [MPa]	G_{theo} [MPa]	% diff
25mm	80923.74	80000	1.15
39mm	79792.92	80000	0.26
41mm	80656.36	80000	0.82

Table 7: Percent difference between calculated and literature values of shear modulus.

As shown in the table above, the percent difference is the largest for the 25mm specimen, which makes sense because it is the shortest specimen.

CONCLUSIONS

This lab successfully achieved its main objective of determining the shear modulus (G), and initial tension (F_i) of helical extension springs using non-linear least-squares regression model and Castigliano's Second Theorem. The calculated values of G are listed in Table 6, where the G values were 80.924 GPa, 79.793 GPa, and 80.656 GPa for the 25 mm, 39 mm, 41 mm springs, respectively. Table 7 shows that all measured G values were within 1.2% error range, confirming strong experimental accuracy. Moreover, the spring rate k was derived from the spring's geometry, showing that the 25 mm spring was the stiffest and the 39 mm spring was the least stiff. The variations in the number of active coils N_a, demonstrated a much expected inverse relationship with stiffness. Lastly, by isolating the linear region of the graph obtained (Figure 1), it can be concluded that the experiment produced consistent and expected results, validating theory as well as the method of experimentation chosen.

The unified graph in Figure 1 shows the relationship between the force applied and the deflection of the spring for the three specimens tested in this lab. To utilize the actual power of this graph to show the way each spring deflects with the force applied, it is important to understand the meaning of

the different regions of the graph. For instance, to identify the range of data values that correspond to the linear trend, one must look for the portion of the curves where the data points align closely in a straight line. This would be an indication of a consistent linear relationship between force and deflection. This behavior is typically seen in the beginning of the curve, where the material exhibits elastic behavior and the spring follows Hooke's Law (F = -kx). To get a visual understanding of this part of the graph, the constant gradient portion of the curve, containing no curvature, flattening, or sudden deviation needs to be considered. For the purpose of this lab, the method chosen to detect this portion more accurately was by plotting the raw data collected and then locating the straight segment after the graph curved, but before it regressed back to the origin, with a negative slope. For example, the 25 mm spring showed a linear range from approximately 0.25 mm to 2.6 mm, where the force increased proportionally with deflection. The main aim behind being precise with this portion of the curve is because this precision determines the accuracy of the measurement of various mechanical properties. These include, the spring rate k, the initial tension F_i,and the shear modulus (G), as these properties can only be determined when the spring is under purely elastic loading.

The estimation of the shear modulus (G) and the initial tension (F_i) of the helical springs was done using a non-linear least squares optimization approach. The force-deflection relationship established from Castigliano's Second Theorem in Equation 2, is non-linear with respect to the above mentioned two parameters. The force-deflection relationship also incorporates the spring geometry, such as the mean coil diameter D, free length L_0, and the wire diameter d. This can be seen in equation 2.

Thus, a standard linear regression model could not be developed and implemented. Instead, the choice of MATLAB's function $fmincon$ was considered the best choice. The function was used to minimize the squared error between the force measurements and the predictions from the model. The optimization routine solved the following:

$$\min_{G>0,\, F_i>0} \sum \left[F_{\text{measured}} - (F_i + k\Delta)\right]^2$$

wherein k is given Equation 3. In this case, the number of active coils N_a was calculated based on the number of coils N_b and the Poisson's ratio ν using Equations 4 and 5 .

In general, all the force and deflection values were taken from the raw data collected. To use the $fmincon$ function, initial guesses were made and bounds were defined to provide logical physical constraints such as $G > 0$. Furthermore, detail was given to properly scale the parameters to avoid any ill-conditioning. This method proved efficient as it matched the theoretical values within a 1.2% error range, confirming the validity of the method used.

The number of active coils, N_a and the spring rate k are important parameters to characterize the mechanical behavior of helical extension springs. Specifically, active coils are those coils that are able to deform elastically under an applied load, contributing to the total deflection of the spring. N_a can be determined by using Equation 4.

In general, as N_a increases, the spring becomes increasingly compliant, as it distributes strain over more coils and thus reduces stiffness k. The spring rate k defines the slope of the force-deflection curve and represents the spring's stiffness. It increases with larger diameter d and shear modulus G. Moreover, it decreases with a larger mean coil diameter D or active coils N_a. Thus, a stiffer spring (with a larger k value) deflects less under a given applied load, making it suitable for applications including bearing or other precision applications. Conversely, a spring with low k offers greater deflection, making it ideal for shock absorbing applications or cushioning systems. As mentioned earlier, the implications of the spring rate and number of active coils was manifold in this lab because it was used in determining the material properties such as G. These parameters, hence, directly influence the accuracy of the derived force-displacement model, and in turn the reliability of the lab's measurements and conclusions.

The calculated shear modulus (G) values for the three helical extension springs investigated in this

lab were found to be in close agreement with the literature values for the same material, steel music wire. For instance, the literature value for G for the steel music wire is approximately 80 GPa. The experimental values obtained, as seen in table 7, were 80.924 GPa (25 mm spring), 79.793 GPa (39 mm), and 80,656 GPa (41 mm spring). The corresponding percentage errors were 1.15% (25 mm spring), 0.26% (39 mm spring), 0.82% (41 mm spring). While these differences are relatively small, they still indicate slight variation among the specimens. The 25 mm spring, exhibited the highest deviation from literature values, which can be attributed to its shorter length and steeper force-deflection curve, making it more sensitive to small misalignments. Whereas, the 39 mm spring the closest match to the literature value, suggesting more stable behavior and a possible reduced noise over its linear region. These minor differences could be due to manufacturing variation, such as residual stresses or even coil winding. These considerations were not made while performing the data reduction for this lab's purpose. However, despite these minor inconsistencies, the overall results reinforce the conclusion that all the three springs match the expected behavior of steel music wire, successfully validating the experimental approach and the assumptions used while developing the force-deflection model.

REFERENCE LIST

[1] BYJU'S. "Shear modulus (modulus of rigidity) - definition, formula, units, examples." (2022), [Online]. Available: `https://byjus.com/physics/shear-modulus-elastic-moduli/` (visited on 03/17/2025).

[2] M. Gonzalez. "Me 32301 - mechanics of materials laboratory." (2023).

[3] MatWeb. "Astm a228 steel (uns k08500)." (), [Online]. Available: `https://www.matweb.com/search/datasheet.aspx?matguid=4bcaab41d4eb43b3824d9de31c2c6849&ckck=1` (visited on 04/08/2025).

[4] OpenAI. "ChatGPT (April 4th, 2025 Version)," [Online]. Available: `https://openai.com/chatgpt(visitedon03/04/2025).`

DATA SHEETS

The following three figures are screenshots of a set of the data collected by the team in lab. The data contains the Displacement (in mm) and Force (in N) for three springs with different lengths. The three springs are 25 mm, 39 mm, 41 mm in length.

	A	B	C
1	Time	Displacement	Force
2	(s)	(mm)	(N)
3	0	0	-0.0225
4	0.02	0	0.0079
5	0.04	0	-0.004
6	0.06	0	-0.0113
7	0.08	0	-0.0012
8	0.1	0	-0.0186
9	0.12	0	0.0195
10	0.14	0	0.0166
11	0.16	0	-0.0222
12	0.18	0	0.0009
13	0.2	0	-0.0158
14	0.22	0	-0.0161
15	0.24	0	-0.0124
16	0.26	0	-0.0093
17	0.28	0	0.004
18	0.3	0	0.026
19	0.32	0	0.0283
20	0.34	0	0.0112
21	0.36	0	-0.0032
22	0.38	0	-0.0286
23	0.4	0	-0.0458
24	0.42	0	-0.0312
25	0.44	0	0.0105
26	0.46	0	0.0259
27	0.48	0	0.017

Figure 2: Datasheet for 25 mm Spring

	A	B	C
1	Time	Displacement	Force
2	(s)	(mm)	(N)
3	0	0	0.0429
4	0.02	0	0.0212
5	0.04	0	-0.023
6	0.06	0	-0.0117
7	0.08	0	-0.0023
8	0.1	0	-0.0283
9	0.12	0	-0.0354
10	0.14	0	-0.0135
11	0.16	0	-0.0126
12	0.18	0	-0.0283
13	0.2	0	0.0207
14	0.22	0	0.0039
15	0.24	0	-0.0377
16	0.26	0	0.0065
17	0.28	0	0.0289
18	0.3	0	-0.0201
19	0.32	0	0.0129
20	0.34	0	0.0031
21	0.36	0	-0.0291
22	0.38	0.0001	-0.0166
23	0.4	0.0001	0.0066
24	0.42	0.0001	0.0076
25	0.44	0.0001	0.0083
26	0.46	0.0001	-0.0007
27	0.48	0.0001	-0.0223

Figure 3: Datasheet for 39 mm Spring

	A	B	C
1	Time	Displacement	Force
2	(s)	(mm)	(N)
3	0	0	-0.0168
4	0.02	0	0.0058
5	0.04	0	-0.0001
6	0.06	0	-0.0154
7	0.08	0	-0.006
8	0.1	0	-0.0082
9	0.12	0	0.0001
10	0.14	0	-0.0177
11	0.16	0	-0.0062
12	0.18	0	-0.0127
13	0.2	0	-0.0117
14	0.22	0	-0.0239
15	0.24	0	-0.0169
16	0.26	0	-0.0236
17	0.28	0	0.002
18	0.3	0	0.0155
19	0.32	0	0.0093
20	0.34	0	0.0159
21	0.36	0	-0.0054
22	0.38	0.0001	-0.0273
23	0.4	0.0001	-0.0062
24	0.42	0.0001	-0.0051
25	0.44	0.0001	-0.0115
26	0.46	0.0001	0
27	0.48	0.0001	-0.0106

Figure 4: Datasheet for 41 mm Spring

APPENDIX

The following code was used to find the boundaries of the linear portions of the data all three raw data sets:

```

%set up data for 25 mm
Filename_1="2025_SP_Section08_Group04_Lab09_25mm1_14";
Filetype_1='.csv';
data_1=readmatrix(append(Filename_1,Filetype_1));

%create 25mm graph
figure (1)
hold on
box on
grid minor
xlabel("Displacement \Delta [mm]")
ylabel("Force F [N]")
title('My Title')
ax = gca;
ax.FontName='Arial';
ax.FontSize=14;
plot(data_1(:,2), data_1(:,3));

%set up data for 39mm
Filename_2="2025_Spring_Section08_Team4_Lab9_39mm1_12";
Filetype_2='.csv';
data_2=readmatrix(append(Filename_2,Filetype_2));

%create 39mm graph
figure (2)
hold on
box on
grid minor
```

```
xlabel("Displacement \Delta [mm]")
ylabel("Force F [N]")
title('My Title')
ax = gca;
ax.FontName='Arial';
ax.FontSize=14;
plot(data_2(:,2), data_2(:,3));

%set up data for 41mm
Filename_3="2025_SP_Section08_Team04_Lab09_41mm1_10";
Filetype_3='.csv';
data_3=readmatrix(append(Filename_3,Filetype_3));

%create 41mm graph
figure (3)
hold on
box on
grid minor
xlabel("Displacement \Delta [mm]")
ylabel("Force F [N]")
ax = gca;
ax.FontName='Arial';
ax.FontSize=14;
plot(data_3(:,2), data_3(:,3));
```

The following code is the function used for the 25 mm spring data set, to find shear modulus G and F_i. Two similar function files were created for the 39 mm and 41 mm spring data respectively:

```

function estimateSpringParameters()

    %set up data
    Filename="2025_SP_Section08_Group04_Lab09_25mm1_14";
```

```
Filetype='.csv';
data=readmatrix(append(Filename,Filetype));

%In the space below, import the data sheet(s) obtained from the Instron UTM.

%set up linear data rows
row_s = find(data(:,2) == 0.25, 1, 'first');
row_f = find(data(:,2) == 2.6, 1, 'first');
row = row_s:row_f;

% Referencing the imported data sheet, define the list of values for
% force applied and vertical deflection.
F = data(row,3); % Measured force values [N]
Delta = data(row,2); % Measured deflection values [mm]

% Define the known values
d = 1; % Wire diameter [mm]
D = 6; % Coil diameter [mm]
nu = 0.313; % Poisson's ratio
L0 = 25; % Free length of the spring [mm]

% Define the objective function (leave)
objfun = @(x) sum((F - x(1)*d^4/(8*D^3*(L0/d+1-2*D/d+1/2/(1+nu)))*Delta - ...
x(2)*ones(size(F))).^2);

% Define the constraints
options = optimoptions(@fmincon,'Algorithm','interior-point');
lb = [0, 0]; % Lower bounds
ub = [Inf, Inf]; % Upper bounds

% Initial guess for G and Fi
```

```
    G0 = 1;
    Fi0 = 0;
    x0 = [G0, Fi0];

    % Call fmincon to minimize the objective function
    x = fmincon(objfun, x0, [], [], [], [], lb, ub, [], options);

    % Display the optimal values of G, and Fi
    disp(['Optimal value of G: ', num2str(x(1)), ' MPa']);
    disp(['Optimal value of Fi: ', num2str(x(2)), ' N']);
end
```

The following code was used to find the N_a, N_b, and k:

```
% Plots:
% code outputs Fi = y-intercept
% from lab manual = calculate k = slope
% one graph have all three raw data as well as the models from y = kx + Fi

% OD = D + d
% D = OD - d

%{
% calculating the slope
L1 = 25 mm; D1 = 6 mm; d1 = 1 mm;
L2 = 39 mm; D2 = 4.7 mm; d2 = 0.8 mm;
L3 = 41 mm; D3 = 7.3 mm; d3 = 1.2 mm;

L= 2(D-d) + (Nb+1)d;
%}

% Given values
L = [25, 39, 41];            % in mm
D = [6, 4.7, 7.3];           % in mm
```

```
d = [1, 0.8, 1.2];          % in mm

% Calculate Nb for each case
Nb = ((L - 2 .* (D - d)) ./ d) - 1;
disp(Nb)

%Caculate Na for each case
nu = 0.313;
Na = Nb + 1 ./ (2 * (1 + nu));

%calcualte slope k
G = [80923.7376, 79792.9201, 80656.3595];   % shear modulus values (MPa)
k = (d.^4 .* G) ./ (8 .* D.^3 .* Na);
```

The following code file was used to compile the raw data of all three springs, as well as plot their linear model trendlines:

```

%set up colors
colorlist=["#6c3483","#CCCCFF", "#6495ED","#40E0D0","#9FE2BF",...
    "#DE3163","#FF7F50","#FFBF00","#f7dc6f","#2b7d38"];

% set up trendline variables
Fi = [2.3181, 3.317, 0.99876]; %y-intercept
k = [3.2565, 1.0252, 2.2985];  %slope

%set up graph parameters
hold on
box on
grid minor
xlabel("Deflection \Delta [mm]")
ylabel("Force F [N]")
ax = gca;
```

```
ax.FontName='Arial';
ax.FontSize=14;

%set up data for 41mm
Filename_3="2025_SP_Section08_Team04_Lab09_41mm1_10";
Filetype_3='.csv';
data_3=readmatrix(append(Filename_3,Filetype_3));
x3 = data_3(:,2);          % Displacement
y3 = k(3) * x3+ Fi(3);     % Trendline

%set up plot for 41mm
plot(x3, data_3(:,3), 'o', 'LineWidth', 1, 'Color', colorlist(1), ...
    'MarkerFaceColor', colorlist(1), 'DisplayName', '41mm Spring Data');

hold on

%set up data for 39 mm
Filename_2="2025_Spring_Section08_Team4_Lab9_39mm1_12";
Filetype_2='.csv';
data_2=readmatrix(append(Filename_2,Filetype_2));
x2 = data_2(:,2);           % Displacement
y2 = k(2) * x2 + Fi(2);     % Trendline

%set up plot for 39mm
plot(x2, data_2(:,3), 'o', 'LineWidth', 1, 'Color', colorlist(3), ...
    'MarkerFaceColor', colorlist(3), 'DisplayName', '39mm Spring Data');

hold on

%set up data for 25mm
Filename_1="2025_SP_Section08_Group04_Lab09_25mm1_14";
Filetype_1='.csv';
data_1=readmatrix(append(Filename_1,Filetype_1));
```

```
x1 = data_1(:,2);          % Displacement
y1 = k(1) * x1+ Fi(1);     % Trendline

%set up plot for 35mm
plot(x1, data_1(:,3), 'o', 'LineWidth', 1, 'Color', colorlist(6), ...
    'MarkerFaceColor', colorlist(6), 'DisplayName', '25mm Spring Data');

%plot trendlines
plot(x3, y3, '-.', 'Color', colorlist(2), 'LineWidth', 2, 'DisplayName', '41mm
    Spring Model');
plot(x2, y2, '-.', 'Color', colorlist(5), 'LineWidth', 2, 'DisplayName', '39mm
    Spring Model');
plot(x1, y1, '-.', 'Color', colorlist(8), 'LineWidth', 2, 'DisplayName', '25mm
    Spring Model');

% Add the legend
legend('Location', 'best');

hold off

exportgraphics(gcf, "Lab9.png");
```